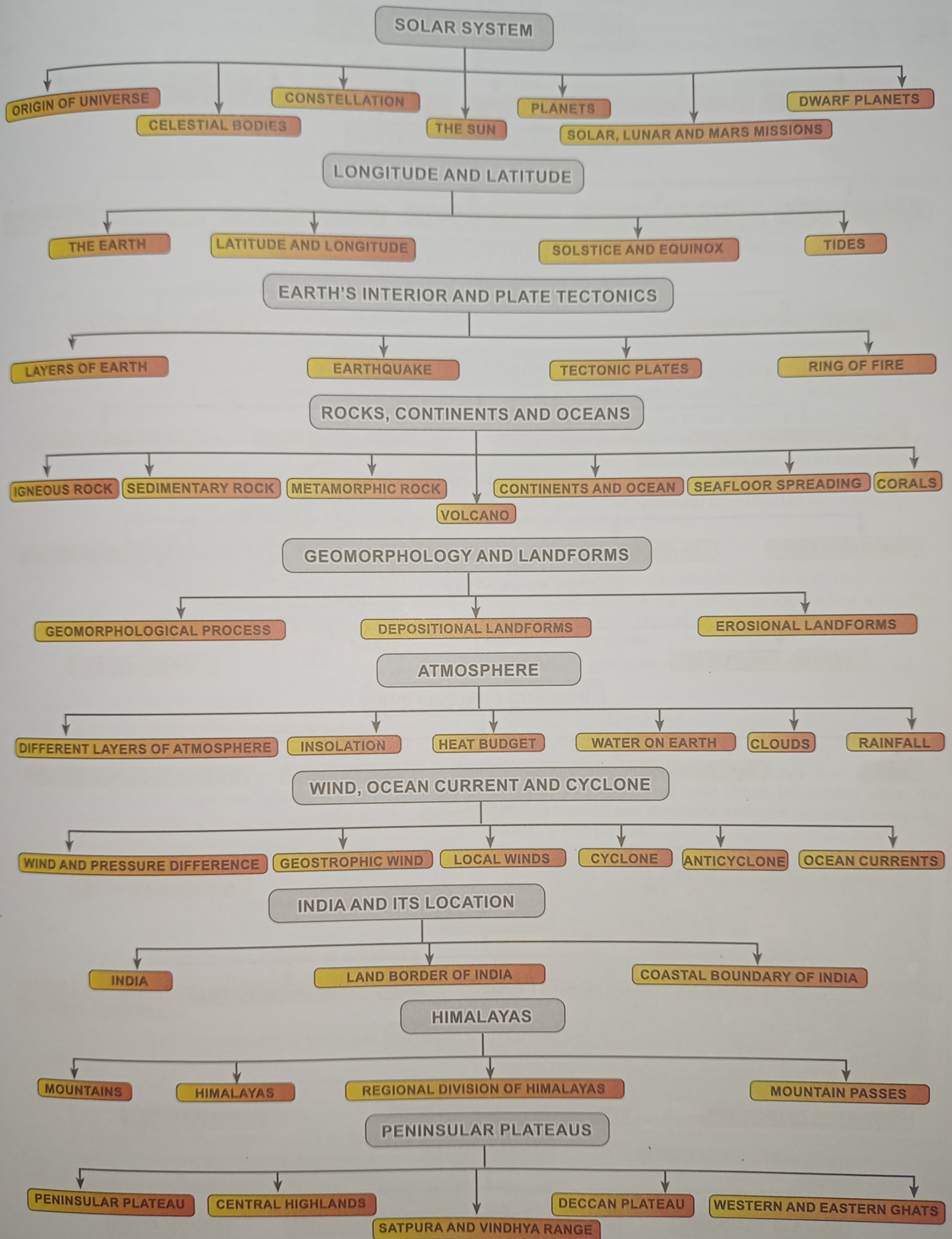
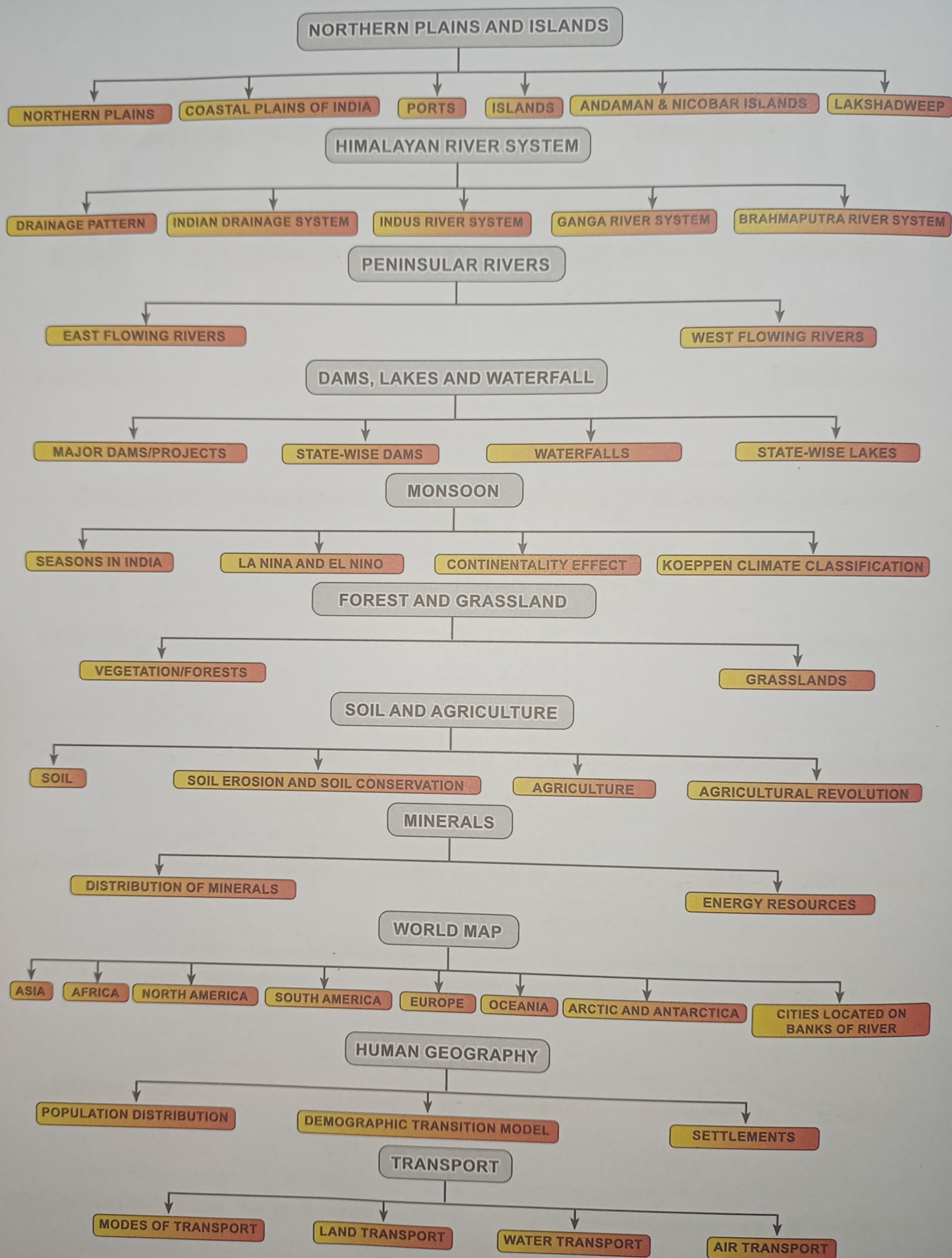


GEOGRAPHY







- Study of Universe is known as: **Cosmology**.
- Universe consists of many **Galaxies**, a galaxy in turn consists of many solar systems.
- Branch of science that deals with celestial bodies: **Astronomy**.

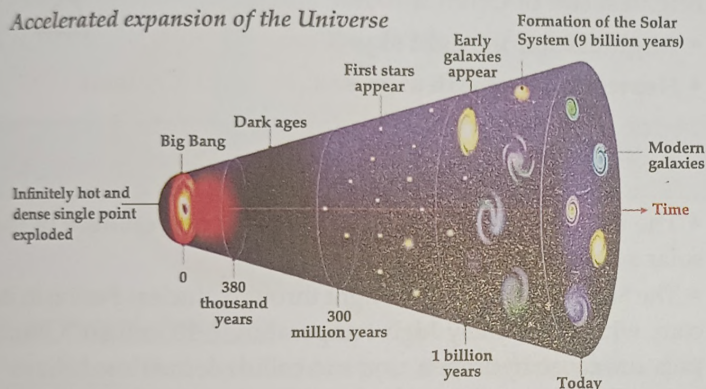
ORIGIN OF UNIVERSE

- Various theories have been given on how the universe came into existence.

Big Bang Theory

- Big Bang was an explosion of concentrated matter in the universe that occurred **13.8 billion years ago**, leading to the formation of galaxies of stars and other heavenly bodies.
- Origin of Big Bang Theory (also known as Big Bang Explosion) was given by **Georges Lemaitre** in **1927** and published in **1931**.

Accelerated expansion of the Universe



Edwin Hubble Theory - (1929)

- Edwin Hubble proposed the expanding universe hypothesis, demonstrating that distant galaxies are moving away from Earth with velocities proportional to their distances (now known as Hubble's Law), which provided the first strong observational evidence that the universe is continuously expanding.

Steady State Theory - (1948)

- It was given by **Fred Hoyle**, Hermann Bondi, and Thomas Gold.
- This theory proposes that the **universe is constantly expanding, but its average properties remain unchanged**.

Galaxy

- A galaxy is a massive collection of stars, gas, dust, and dark matter held together by gravity.
- Our galaxy is **Milky Way Galaxy (or the Akash Ganga/ Mandakini)** formed after the Big Bang.
- **Andromeda/M-31** is the nearest galaxy to the Milky Way.

Our solar system lies in the Orion Spur, a small spiral arm of the Milky Way located between the Sagittarius and Perseus arms.

ORIGIN OF SOLAR SYSTEM

- The formation of Solar System began **4.6 Billion years ago**, while the age of Earth is **4.5 Billion years** old.

Theories related to the origin of solar system

1. Copernicus Theory:

- De revolutionibus orbium coelestium is the Heliocentric theory of **Nicolaus Copernicus**, given in **1543**.
- It predicted that the **Sun is at the centre** of the universe, with the Earth and other planets orbiting around it.

2. Nebular Hypothesis:

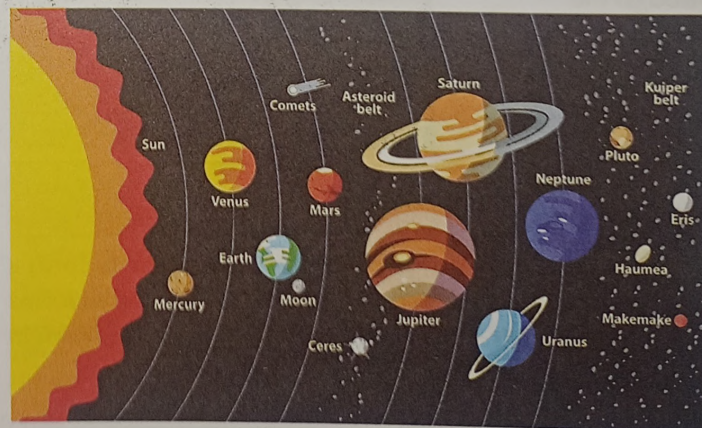
- First proposed by **Immanuel Kant** in **1755** and independently expanded upon by **Pierre-Simon Laplace** in **1796**.
- The process started when a cloud of gas and dust collapsed, forming a spinning disk called the solar **Nebula**.
- Gravity pulled material to the center, forming the Sun, while the rest flattened into a spinning disk. Dust and gas clumped into planetesimals, which merged to form planets. The Sun formed in the center, and the planets formed in a thin disk orbiting around it → **It is the most widely accepted hypothesis**.

3. Planetesimal Theory:

- **Thomas Chrowder Chamberlin** and **Forest Ray Moulton** concluded (in **1905**) that a passing star came close to the Sun and, due to its gravitational attraction, a portion of the Sun's matter was drawn out.
- This matter gradually cooled and combined to form the planets.

4. Binary Star Hypothesis:

- It was proposed by **H.N. Russell** in **1937**.
- The Binary Star Hypothesis proposes that Earth and the other planets formed from debris left behind when the Sun's companion star exploded, with the material eventually coalescing into planets.



- Our Solar System has **8 planets** and **5 Dwarf planets**.

CELESTIAL BODIES

- Celestial bodies are objects in space that glow in the night sky, such as planets, stars, moons, asteroids, comets, and galaxies.

They are classified into

Luminous

Those celestial bodies that emit their own light.

Eg. Stars, Firefly

Non-luminous

Those celestial bodies that do not emit their own light.

Eg. Moon, Comets, Asteroids, Meteorites, Meteor, Meteoroid.

Comets

- Comets are **large, icy objects** that orbit the Sun. They are made of dust, rock, and frozen gases like water, carbon dioxide, ammonia, and methane and when it passes near the Sun, its ice starts melting, creating two tails: ion tail & dust tail.
- Comets generally come from the Kuiper belt (short-period comet) or the Oort cloud (long-period comet).

Oort Cloud: Cloud-like form of icy objects and dust particles, beyond the Kuiper belt.

- Halley's Comet:** It is **seen every 76 years**. It was last seen in 1986 and is expected to return in 2061.

Asteroid

- An asteroid is a **rocky, metallic, or icy object** that orbits the Sun.

Meteoroid

- A meteoroid is a **small, rocky or metallic space rock** that orbits the sun. These are **much smaller than asteroids**, ranging in size from dust grains to objects up to a meter wide.

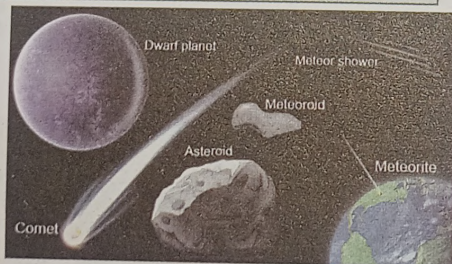
Meteor

- When a meteoroid enters Earth's atmosphere (in Mesosphere), **it burns up and creates a streak of light** called a meteor (also known as **Shooting Star**).

- Meteor shower name Quadrantids is visible in the early month of January.**

Meteorite

- If the meteoroid survives its trip through the atmosphere, it hits the ground and is called a meteorite.



CONSTELLATION

- A constellation is a group of stars that appear to form a pattern or shape in the night sky. Every constellation forms an asterism which is a group of stars that form a pattern in the night sky.

Ursa Major Constellation

- It is also known as the **Great Bear and Saptarishi (in India)**, is a constellation in the northern sky. Big Dipper asterism is seen in Ursa Major (**Brightest star - Alioth**).



Ursa Minor

- Ursa Minor, also known as the Little Bear, is a prominent constellation in the northern sky (**Brightest star - Polaris**).

Cassiopeia

- It is a W-shaped constellation. **The brightest star of Cassiopeia is Schedar** (also known as **Alpha Cassiopeia**).

Cygnus

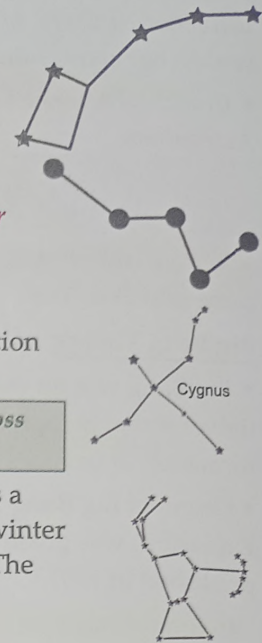
- It is a prominent northern constellation known for its distinctive **cross shape**.

Sagittarius: Bow and cross shaped constellation.

Orion

- Orion (also known as **The Hunter**) is a prominent set of stars visible during winter in the northern celestial hemisphere. The brightest star of Orion is **Rigel**.

- Brightest star in night sky: Sirius.**
- Nearest star to Earth after sun: Proxima Centauri.**



THE SUN

- The only star in our solar system and **powerhouse** of the solar system.
- The Sun produces its own light through Nuclear Fusion in its core, where extremely high temperature (~15 million °C) and pressure cause hydrogen atoms to collide & fuse into helium.
- It is composed of **Hydrogen (73%), Helium (25-28%) and other gases**.
- It carries almost **99%** of the mass of our solar system.

The science of studying the Sun and its influence throughout the solar system is called Heliophysics.

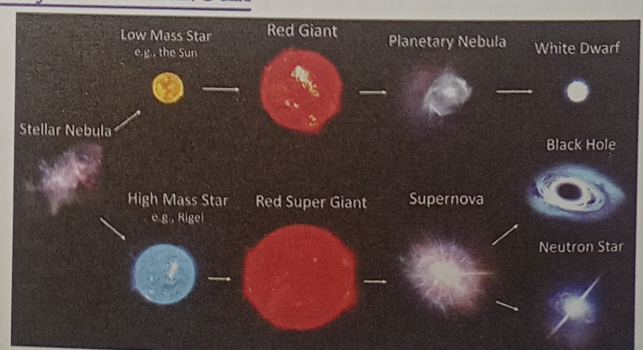
- Approximately **15 crore Kilometre (149.6 Million Km)** further away from Earth. It takes around **8 minutes 20 seconds** for light at the speed of **3 lakh Km/sec** to reach the earth.

- Temperature at surface = 5800K or 5600°C**
- Temperature at the center = 15.7 million K**

Note:

*K: Kelvin

Life cycle of a star/Sun

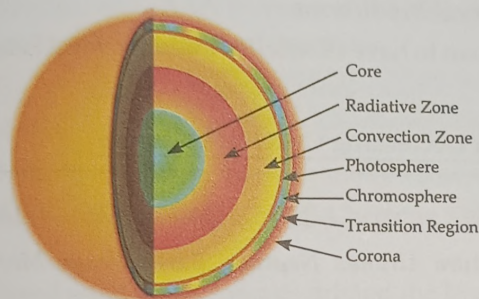


- The life cycle of a star, including our Sun, is determined primarily by its mass: low-mass stars live for tens of billions of years, medium-mass stars (like the Sun) for about 10-12

billion years. Eventually **small stars become white dwarfs**, whereas **massive stars end up** as a neutron star or a **black hole** following a fierce supernova explosion.

- **Stellar Nebula:** Birthplace of stars, clouds of gas and dust.
- **Average Star:** Current stage of the Sun (~4.6 billion years old, halfway through its ~10 billion-year life).
- **Red Giant:** When hydrogen runs out, the Sun will expand enormously and shed its outer layers.
- **Planetary Nebula:** Outer layers drift away, leaving the hot core.
- **White Dwarf:** Dense, faint remnant of the Sun's core.
- **Black Dwarf:** Eventually cools and fades into a cold, dark remnant.

Layers of Sun: Core < Radiative Zone < Convective Zone < Photosphere (Outermost visible part) < Chromosphere < Corona (Outermost layer but not visible with naked eye, extremely hot and only visible during solar eclipses).



Solar flares & Sunspots

Solar flares: These are sudden bursts of radiation caused by magnetic activity on the Sun (short-lived powerful bursts of energy).

Solar wind: It is the continuous flow of charged particles from the Sun's corona (steady, ongoing flow of particles).

Sunspots: These are temporary, planet-sized, darker regions on the Sun's surface caused by intense magnetic activity that inhibits heat transfer, making them cooler than the surrounding photosphere.

Magnetic Cycle: The Sun's magnetic activity peaks every 11 years, increasing the number of sunspots and solar flares.

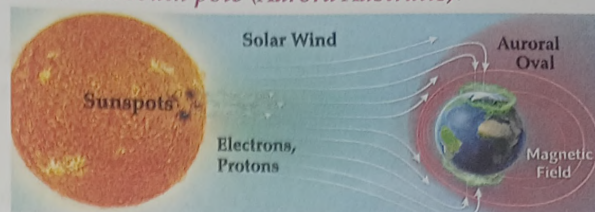
Solar flares effect on Earth: Damage spacecraft electronics, GPS signal disturbance, Geomagnetically induced current in power systems which may cause black-outs, Radiation effects on avionics and communication, induced effects in submarine cables etc.

When solar flares and solar winds interact with Earth's magnetic field, they trap charged particles, forming the Van Allen Radiation Belt around Earth.

Difference between Solar flares and Coronal Mass Ejections (CMEs):

Both phenomena are caused by magnetic reconnection on the Sun, but CMEs involve the expulsion of physical material, whereas solar flares are primarily bursts of light and high-energy particles. While they often occur together, they can also happen independently.

Aurora: As the magnetic field of Earth is lesser in the polar regions, Sun's solar flares hit the atmosphere on poles, creating such light phenomenon in both **north pole (Aurora Borealis)** and **south pole (Aurora Australis)**.



PLANETS

Classification of Planets

Terrestrial Planets	Jovian Planets
They are relatively very small	They are immense in size
They are made of rocky material	They are made of gaseous material
Their surface is solid	They do not have a solid surface
They are nearer to Sun	They are far away from Sun
They have few /no moons	They have multiple moons
They do not have rings	They support ring systems
Eg. Mercury, Venus, Earth, Mars	Eg. Jupiter, Saturn, Uranus, Neptune

Mercury

- It is the **closest planet** to the Sun.
- It is the smallest planet in the solar system with a diameter of **4900Km**.
- It is the fastest planet with speed of **172500 km/h** or **47 km/sec** to complete revolution around Sun in **88 days**.
- It is the only planet that has **no atmosphere**.
- It is a planet with second highest density

Jupiter and Saturn (gas giants),
Uranus and Neptune (ice giants)

Venus

- It is the **hottest planet** in the solar system.
- It is also known as "**Earth's Twin**". It is because of the similarity in size and mass between Venus and Earth.
- **Brightest planet in the Solar system**. It can be seen in the morning and evening with open eyes.
- This planet has the **slowest rotation**. It takes **243 Earth days** to complete one rotation.
- Venus is covered with thick clouds of **Sulfuric acid (H₂SO₄)**.
- **Other names for Venus:** Morning star, Evening star, Lucifer (light bringer).

- All planets have satellites except Mercury and Venus
- All planets rotate from West to East except Venus and Uranus (they rotate from East to West)

Earth

- The only Planet to give support to life with a pleasant atmosphere.
- It is also known as "**Blue Planet**" because of the presence of water (**70%**) on it.

SOLAR SYSTEM

- It has *one natural satellite* named "Moon".
- It is a planet with *highest density* (5.51 g/cm^3)
- Closest planet to Earth is Venus.

Mars

- It is known as "*Red Planet*" because of iron-rich red soil.
- It is the *second smallest planet* in the solar system after Mercury.
- It has two natural moons "*Phobos*" and "*Deimos*".
- It has a thin atmosphere and surface covered with valleys, craters, deserts, and ice caps etc.
- "*Olympus Mons*" (height: 22–25 km) is the largest volcano and the tallest mountain in the solar system and lies on Mars.
- Length of a day in Mars equal to *24 hr and 37 minutes*.
- The *asteroid belt* is found between the orbits of Mars and Jupiter.
- Asteroid Benu studied by NASA's OSIRIS-REx mission.

Jupiter

- It is the largest planet of the solar system with *the shortest rotation* of 9 hours 55 minutes.
- It has an atmosphere filled with Hydrogen, Helium and other gases.
- Jupiter is the *third brightest object in the night sky* after the Moon and Venus, but generally three brightest objects in the sky are Sun, Moon and Venus.
- Great Red Spot, a giant storm in the solar system, exists on this planet.
- At present it has *97 moons*, including four large Moons "*Io, Europa, Ganymede, and Callisto* (has evidence of *Ozone layer*)", also known as *Galilean Moons* because they were discovered by *Galileo*. "*Ganymede*" is the largest among them.
- It has a faint ring system.

Saturn

- It is the *second largest planet* in the solar system.
- It has bright and concentric rings around it which are made up of tiny rocks and pieces of ice.
- Saturn is the planet with *the least density* (0.69 g/cm^3). Apparently, it can float in water.
- At present it has *274 moons* and *Titan* is the largest among them.
- Satellites of Saturn → *Titan* (largest moon) and *Enceladus* show possibility of life on it.
- In 1655: Huygens discovered Saturn's rings.
- In 1675: Cassini discovered the gap between rings.
- Rotation time- *10.7 hours*.
- Revolution time- *29.4 Earth years*.

Uranus

- It is greenish in colour, hence known as the "*Green Planet*" due to the presence of *Methane gas* in its atmosphere (CH_4).
- It was discovered by *William Herschel* in 1781.
- The atmosphere of Uranus is composed of Hydrogen and Helium primarily, but it also contains more water, ammonia etc.

- It has *coldest planetary atmosphere* in the solar system.
- It has *at least 27 moons*. Famous moons → *Miranda, Ariel, and Umbriel*.
- It is tilted to 98° at the axis, hence called the "*Rolling/ Lopsided Planet*".
- Rotation time- *17 hours*.
- Revolution time- *84 Earth years*.

Neptune

- It is the farthest planet from the Sun with the *longest/ slowest revolution* of *165 years*.
- Its rotation time is *16 hours*.
- Atmosphere primarily composed of Hydrogen and Helium. It is the windiest planet.
- *It is bluish in colour because of Methane*.
- It was discovered by *Johann Galle and Urbain Le Verrier* in 1846. The only planet in the solar system found by Mathematical Predictions.
- It is known to have *14 satellites* and the most famous moon is *Triton*.

Planets in Decreasing Order of their size

J S U N E V M M
 Jupiter Saturn Uranus Neptune Earth Venus Mars Mercury

Moon

- The Moon is Earth's *only natural satellite*.
- It is *non-luminous in nature*.
- Radii = $1.74 \times 10^6 \text{ m}$
- Moon's light takes *1.26 seconds* to reach Earth's surface.
- Distance between Earth and Moon = *3,84,00km*
- Gravity on Moon = *Earth's Gravity/6*
- Only one side (*near side*) of the Moon is visible (roughly of it is visible) and the remaining side (*far side*) is not visible to the naked eye.

Reason

Rotation time = Revolution time → 27.3 Days

- Highest point on moon: *Mt. Leibniz* (Now officially known as *Mons Huygens*).

SOLAR, LUNAR AND MARS MISSIONS

Sun

- *Aditya-L1 Mission*: This is India's first solar observatory mission launched by the *Indian Space Research Organisation (ISRO)* to study the Sun.

L_1 refers to the first *Lagrange Point* on the Sun. There are a total of *5 Lagrange Points* (L_1 to L_5).

Headquarter of IAU: Paris, France

MoonChandrayaan-3:

• It is the third lunar-exploration mission of Indian Space Research Organisation (ISRO).

• **Lander:** Vikram (landed on 23rd August, we celebrate it as National Space Day).

• **Rover:** Pragyaan.

• **Point:** Shiv Shakti Point.

• Chandrayaan-3 landed

on the **near side of the moon (Lunar South Pole)**.

• **Launch Vehicle:** GSLV-Mk3

Upcoming missions-

Chandrayaan-4: (mission by ISRO) will carry the lunar samples (regoliths) from Moon back to Earth.

Chandrayaan-5: LUPEX (Lunar Polar Exploration) Mission by ISRO (lander) and JAXA (rover and vehicle), explore the southern pole on moon for a longer time.

Artemis II:

Agency: NASA

- Artemis I- tested systems,
- Artemis III- will land humans on the Moon.

Launch Date: April 1, 2026, from Kennedy Space Center.

Crew: Reid Wiseman, Victor Glover, Christina Koch, Jeremy Hansen.

• First crewed mission; astronauts orbited the Moon for 10 days, studied lunar features, and tested life-support systems.

Mars

Mars Orbiter Mission (MOM): (Mangalyaan) is a space probe sent by ISRO, orbiting Mars since 24 September 2014. With this ISRO became the first agency in the world to land on Mars successfully on its first attempt, without any country's help.

Upcoming mission-

MOM-2 is expected to carry a rover and even a pioneering helicopter, making it India's most ambitious Mars project yet.

DWARF PLANETSKuiper Belt

• It is a **ring of bodies beyond Neptune** containing a number of asteroids, rocks, and comets.

Pluto

• As per the new definition of Planets determined by International Astronomical Union (IAU), **Pluto has been omitted from the list of planets in 2006** as it was not be able to clear its orbit of debris.

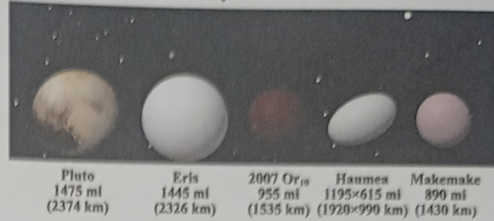
• Pluto is considered as a dwarf planet (size between planets and asteroids) now and it is now a member of **Kuiper Belt**.

• Pluto's largest moon: **Charon (out of 5 moons of Pluto)**.

Pluto was discovered by Clyde Tombaugh.

• **1 revolution of Pluto = 248 Earth years.**

• Pluto's oblong orbit overlaps with that of Neptune.

Largest Dwarf Planet

• **Pluto:** Largest and brightest dwarf planet.

• **Eris:** Second largest dwarf planet.

• **Makemake:** Third largest and it orbits in 310 years.

• **Haumea:** Fastest rotating dwarf planet in Kuiper Belt.

• **Ceres:** This dwarf planet is present in Asteroid Belt.

} Located
in
kuiper
Belt

Three Characteristics of Planets

1. To be around the Sun.
2. To have a nearly spherical shape.
3. They should be able to clear their orbit of debris

The International Space Station (ISS)

• It is the largest man-made object in Space and a collaborative project of 5 space agencies: NASA, Roscosmos, JAXA, ESA, and CSA.

• The first module of the ISS Zarya control module was launched in 1998 & the station was fully assembled in 2011.

• The ISS is in orbit about 400 kilometre above Earth (at an inclination of 51.6°) and travels around Earth at a speed of about 28000 Km/h. This means that it orbits Earth about every **90 minutes**.

India aims to build Bhartiya Antariksh Station by 2035 in a 400 km Orbit above the Earth.

Tiangong is China's self-built space station.

The Chandrasekhar limit

• Proposed by **Subrahmanyan Chandrasekhar** in 1930, earning him the **1983 Nobel Prize in Physics**.

• It is the maximum mass a white dwarf star can have before collapsing under its own gravity—about 1.44 times the mass of the Sun. Beyond this limit, electron degeneracy pressure can no longer resist gravitational collapse, leading to a supernova and the formation of either a neutron star or a black hole.

Terms to Understands

- **Supernova:** Explosion of star.
- **Pulsar:** A neutron star.
- **Black Hole:** A place in space with immense gravity.
- **Van Allen Radiation Belts:** These are two concentric rings of charged particles that surround the Earth.



THE EARTH

• **Geodesy** is the science of accurately measuring and understanding the Earth's geometric shape, orientation in space, and gravity field.

Shape of Earth

• Shape of Earth is **Geoid or Oblate Spheroid** (a little flat from top and bottom) and bulged at the centre.

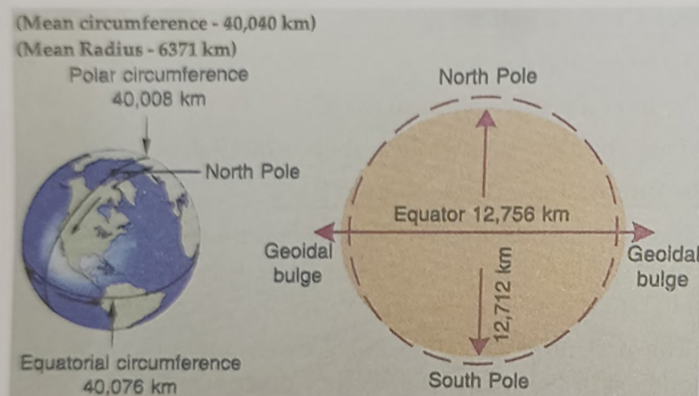
Reason: Centrifugal force (due to rotation of Earth)

• Equatorial radii (6378 km) > Polar radii (6357 km).

Difference between ER and PR is about 21-22 km.

• Equatorial circumference > Polar circumference.

(All because the Earth is bulged at the equator and fattened at the poles).



Rotation of Earth

• Earth rotates on its own axis, from West to East (Anti-clockwise).

• One rotation of Earth = 23 hour 56 minutes 4 seconds.

• Rotational speed is maximum at the equator (Equatorial Speed) and minimum at the poles (Polar Speed).

• **Day and night are caused by the Earth's rotation on its axis.**

Revolution of Earth

• Earth revolves around the Sun in an elliptical orbit.

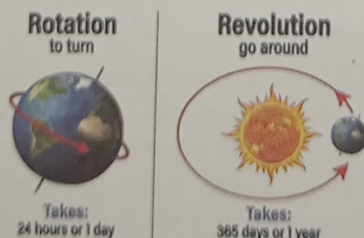
• One revolution = 365 days 5 hours 59 minutes and 16 seconds.

• Orbital speed = 29.8 km/s.

• Mercury has the maximum orbital speed 47.87 km/s at perihelion.

• Neptune has the minimum orbital speed 5.43 km/s.

• **The Earth's revolution around the sun and the tilt of its axis (23.5°) together cause the seasons.**



Rotation vs Revolution

Distance of Earth from Sun and Moon

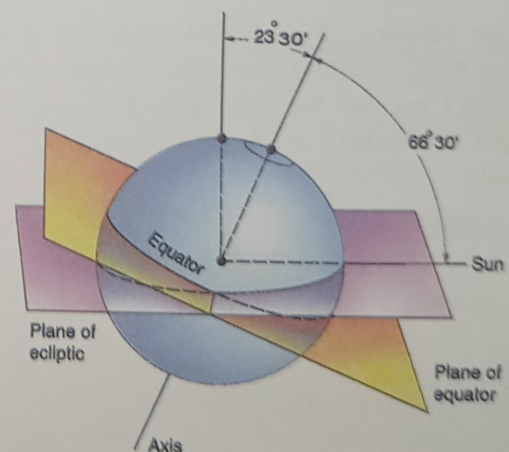
Sun	Moon
- The point in a planet's orbit when it is closest to the Sun is called Perihelion. On January 3rd, the Earth is nearest to the Sun (147 million km). - The point in a planet's orbit when it is farthest from the Sun is called Aphelion. On July 4th, the Earth is farthest from Sun (152 million km).	The point in the moon's orbit when it is closest to Earth is called Perigee. At perigee, the moon appears larger in the sky. The point in the moon's orbit when it's farthest from Earth is called Apogee. At apogee, the moon's tidal-raising force is smaller, and the tidal ranges are less than average.



Inclination of Earth's Axis

• **Axial inclination:** Inclination of Earth on its axis = 23.5°.

• **Orbital inclination:** Inclination of Earth on its orbital plane = 66.5°.



LATITUDE AND LONGITUDE

Latitude

• They are imaginary horizontal lines on the globe that run from **East to West**.

• It is the angular distance of a place from the equator.

• The number of total latitudes is 181 but latitude circles are 179 in number (as 90°N & 90°S are only points, not circles)

• Distance between each latitude is same.

• **1° of latitude = 111 km.**

Important Latitudes

• 0°: Equator (divides Earth into two equal parts) → Northern and Southern Hemisphere.

• $23\frac{1}{2}^{\circ}$ N: Tropic of Cancer.

• $66\frac{1}{2}^{\circ}$ N: Arctic Circle.

• $23\frac{1}{2}^{\circ}$ S: Tropic of Capricorn.

• $66\frac{1}{2}^{\circ}$ S: Antarctic Circle.

• All parallel circles from the equator to the poles are known as "Parallels of Latitude".

• Largest latitude: Equator.
• Smallest latitude: Poles (North and South).

Longitude

• They are *imaginary vertical lines* (also called Meridians) over the globe that run North to South.

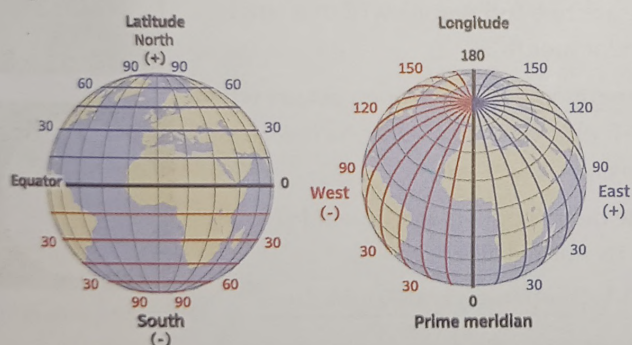
• It is an angular distance of a place from Prime Meridian.

• Distance between each longitude decreases from Equator towards poles, at the poles it is 0, while it is maximum at the equator (111.32 Km), but length of each longitude is same.

• Total longitudes 360°.

Gore: A sector of curved surface of earth that lies between two longitudes, in simple words, Gore is the distance between two consecutive longitudes.

Note - We need both latitude and longitude for the location of a place.



Important Meridians (Longitude)

• Prime Meridian-0° longitude (passes through Greenwich, London).

• 180°E & 180°W are the same longitude (unlike latitudes), that's the opposite of 0° or prime meridian. This 180° meridian is the International Date Line.

• Prime Meridian also known as *International Meridian or Greenwich Meridian*. It passes through

Burkina Faso → B

Spain → S

Ghana → G

Algeria → A

Mali → Me

Togo → In TOGO

UK → KINGDOM

TRICK TO REMEMBER

BSF GAME in TOGO
KINGDOM



Prime Meridian

Concept of Time using Longitude

• 360° = 24 hours

• 15° = 1 hour

• $1^{\circ} = \frac{60^{\circ}}{15} = 4 \text{ minutes}$
(Time taken in each 1°)



International Date Line

Question - If it is 12 noon at 60°E longitude, then explain what would be the time at 30°E Longitude?

Answer. Difference in longitude: 60°E - 30°E = 30°

Time difference: 30° × 4 minutes = 120 minutes = 2 hours

Since 30°E is west of 60°E, it will be earlier in time.

• Moving *East away from Prime Meridian*, the time will *increase* by an *hour* for *every 15°*, consecutively if we *move to West* from the *Prime Meridian*, the time will *decrease* by an *hour* for *every 15°*.

• Entire earth is divided into *24 Time Zones*. *France* has the *most number* of Time Zones (*13*) *Russia* (11 time zones), *US* (6 time zones).

International Date Line (180 E/W)

• It is the *imaginary zig-zag line* passing through the Pacific Ocean from where the date changes.

• While crossing International Date Line eastbound i.e. from *East to West* (Eg. *Japan to USA*) → Date is subtracted by 1 or in other words, *the person gains a day* and vice versa.

Concept of Great Circle

• A great circle is the largest possible circle that can be drawn around a sphere.

• It divides the Earth into *two equal parts called Hemispheres* (Northern and Southern).

• *All longitudes are considered great circles*, but only one latitude is considered a great circle and that is *Equator* (0° latitude).

• *Isochrones: They are lines joining points located at equal travel time from a common centre.*

Hemisphere

• Equal division of Earth in two parts.

• *Equator*: It divides the globe horizontally into two equal parts: *Northern and Southern hemisphere*.

• *Prime Meridian*: It divides the globe vertically in two equal parts: *Eastern and Western Hemisphere*.

• India is located in the Northern part of Equator and Eastern part of Prime Meridian, so it's in the North Eastern hemisphere.



SOLSTICE AND EQUINOX

• **Solstice:** A solstice is an astronomical event that occurs twice a year when the sun reaches its highest or lowest point in the sky, resulting in the longest day or shortest day of the year.

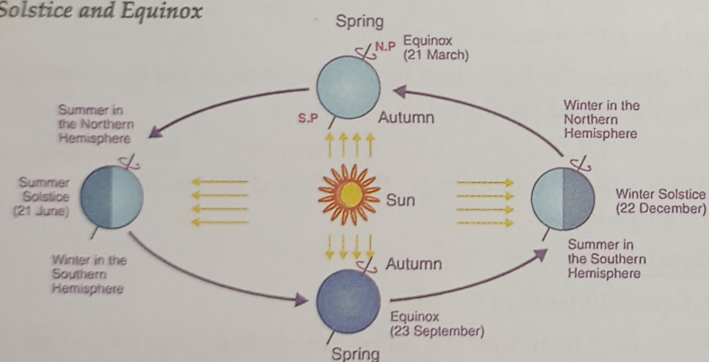
Summer Solstice	Winter Solstice
It occurs on 21st June	It occurs on 22nd December
It is the longest day of the year in Northern Hemisphere	It is the shortest day and longest night of the year in Northern Hemisphere
The vertical rays of Sun are directly overhead on Tropic of Cancer	The vertical rays of Sun are directly overhead on Tropic of Capricorn
Northern Hemisphere gets more heat	Southern Hemisphere gets more heat
Continuous sun rays fall on the North Pole for 6 months	Continuous sun rays fall on the South Pole for 6 months

• **Equinox:** An equinox is a time when the **Sun's apparent path in the sky crosses the equator, making day and night approximately equal in length.** At this position, neither of the poles is tilted towards the Sun.

• **Vernal or Spring equinox:** **21st March.**

• **Autumnal or Fall equinox:** **23rd September.**

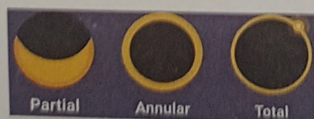
Solstice and Equinox



• **Syzygy:** When the Sun, Moon and Earth, all are aligned in a straight line, the alignment is known as Syzygy.

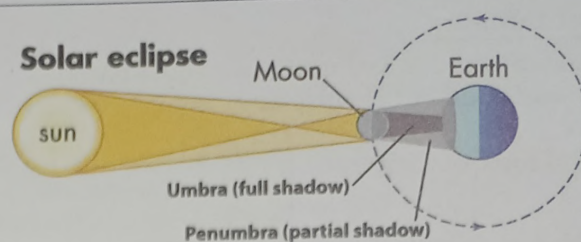
• **Solar eclipse:** It occurs when the **Moon passes between the Earth and the Sun (Conjunction Syzygy)**, casting a shadow on Earth. This can only happen during a **new moon (Amavasya).**

• **Types of Solar Eclipse:** Partial, Annular, Total

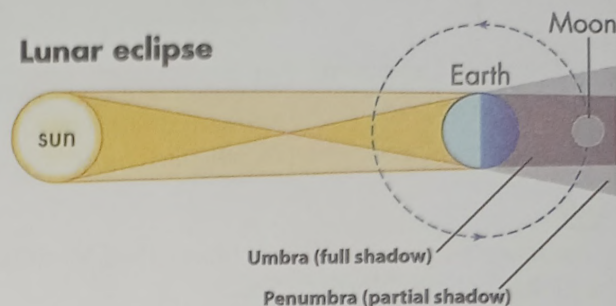


• **Lunar eclipse:** It occurs when the **Earth passes between the Moon and the Sun (Opposition Syzygy)**, casting a shadow on Moon. This can only happen during a **full moon (Purnima).**

Solar eclipse



Lunar eclipse



• **Umbra & Penumbra:** During a solar or lunar eclipse, the darkest, innermost central part of shadow is called Umbra, and lighter, diffused outer part of the shadow that surrounds the umbra, is known as Penumbra.

• **Red moon:** When Earth blocks the Sun's light (light refraction), it causes **blue colour (scatters more)** light to vanish and **red light to reach moon (scatters less).**

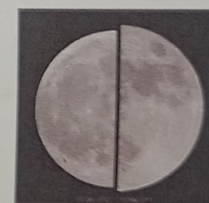


• **Blue moon:** Blue moon (occurs when there are two full moons within a single calendar month).



• **Supermoon:** A supermoon occurs when the **Moon's orbit is closest to the Earth during a full/new moon.**

• The moon appears **larger and brighter** than its normal size.



Torrid, Temperate, and Frigid Zone

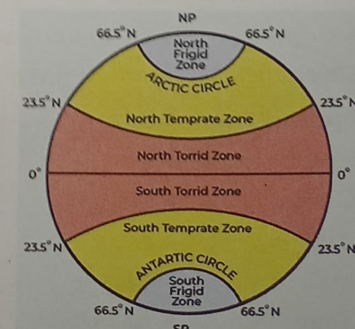
Perigee + Lunar Eclipse

• **Torrid Zone:** Receives direct overhead rays of the Sun due to which it is the **hottest zone.** It is located between the Tropics.

• **Temperate Zone:** Region between Tropic of Cancer & Arctic Circle or Tropic of Capricorn & Antarctic Circle. It receives slanting rays of the Sun.

• **Frigid Zone:** **Coldest region** on Earth which experiences daylight for 6 months and 6 months of night (due to the tilt of Earth). Located beyond the Arctic $\left(66\frac{1}{2}^{\circ}\text{N}\right)$ &

Antarctic Circle $\left(66\frac{1}{2}^{\circ}\text{S}\right)$.



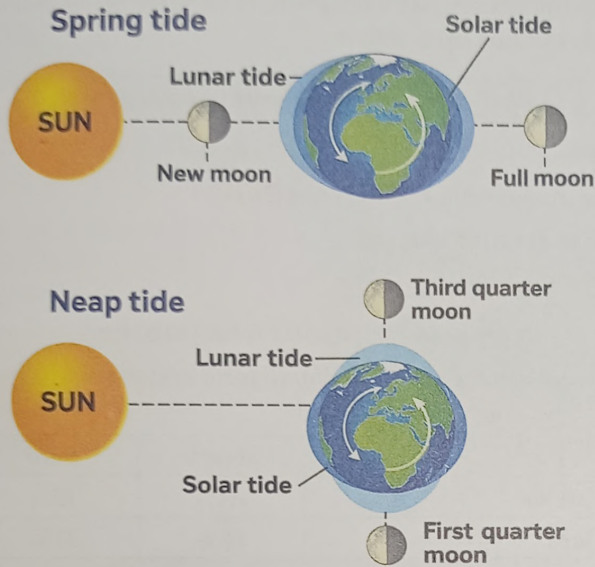
Frigid, Temperate and Torrid Zone

TIDES

• **Tides:** Tides are the periodic rise and fall of the ocean's water levels. They are caused by the gravitational pull of the moon and the sun on the Earth's oceans. They are of two types:

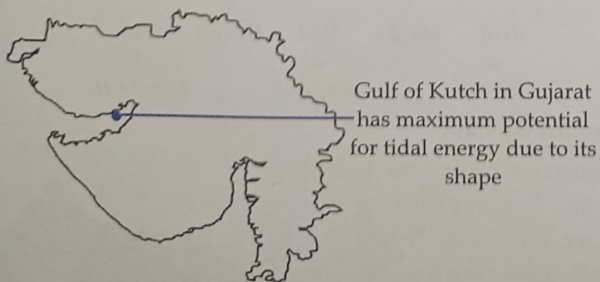
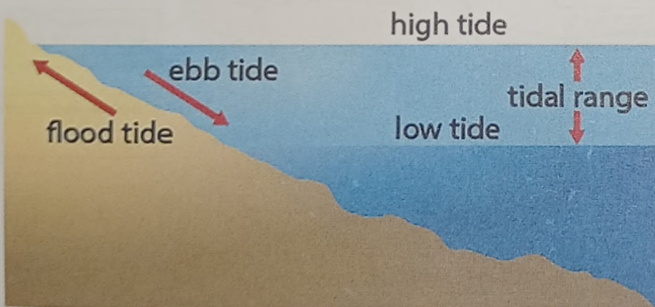
• **Neap Tides:** They occur when Earth, Moon, and Sun are in *perpendicular alignment (Quadrature)*.

• **Spring Tides:** They occur when the Earth, Moon, and Sun are in *straight line alignment (Syzygy)*.



Flood Tide and Ebb Tide

• Flood tide is the *incoming, rising tide that moves toward the coast*, increasing the water level to high tide. Conversely, ebb tide is the outgoing, *falling tide that recedes from the land*, lowering the water level to low tide.



• **Diurnal tides:** Tidal cycles with *one high tide and one low tide each lunar day*, commonly found in the Gulf of Mexico and parts of Alaska.

• **Semi-Diurnal tides:** Tidal cycles with *two nearly equal high tides and low tides every lunar day*.

• The difference of time between two successive High tide and low tide is approximately *6 hours and 12 minutes*.

• The difference of time between two successive high tides is approximately *12 hours and 26 minutes*.



EARTH'S INTERIOR

- Geology:** The scientific study of the Earth, including its composition, structure, rocks, soil, and the processes that shape.

Father of Geology: James Hutton

Sources to know about Earth's interior

Direct Sources	Indirect Sources
Volcanoes	Temperature
Rocks	Meteorites
	Earthquake

- The Kola Superdeep Borehole, located on Russia's Kola Peninsula, is the deepest drilling done on Earth by humans till now, for scientific research, reaching a maximum vertical depth of 12 km.

- Even the mines in South Africa have a maximum depth of 3-4 km.

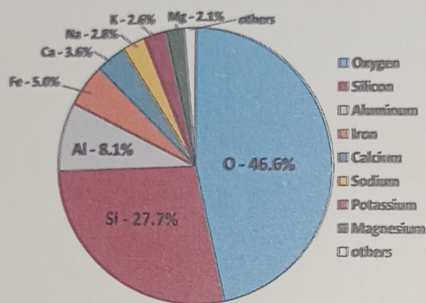
Layers of Earth

Crust

- Generally, the crust is made of **Silica** and **Aluminium** (SiAl).

- Thickness:** 5-70 km.

- The overall crust is brittle in nature.

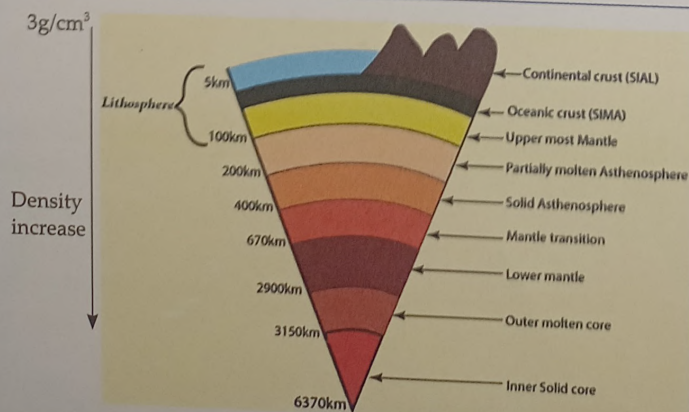


Composition of Earth's crust

Most Abundant Element: Oxygen
Most Abundant Metal: Aluminium

- Crust has two divisions:**

Continental Crust	Oceanic Crust
It is the land of the crust, the outermost layer of the Earth that makes up the continents and continental shelves	It is the water part of the crust, the solid layer Earth's crust that lies beneath the ocean.
It is thick (30 km)	It is thin (5km)
It is lighter	It is denser
It is made of granitic rock	It is made of Basaltic rock.
Mean density - 2.7 g/cm ³	Mean density - 2.9 g/cm ³



Mantle

- It is made of **Silica** and **Magnesium** (SiMa).
- Thickness:** 2900 km.
- Mean density:** 4.5 g/cm³
- Mantle has two divisions:** **Upper Mantle** and **Lower Mantle**.
- Asthenosphere:** It is a **semi-solid/semi-molten layer** of earth's upper mantle that is mechanically weak and ductile. It lies **beneath the lithosphere** (the rigid outermost layer of earth made up of crust and upper solid mantle with 10-200 km of thickness). Asthenosphere is a source of **magma/lava**, which extends upto 400 km.

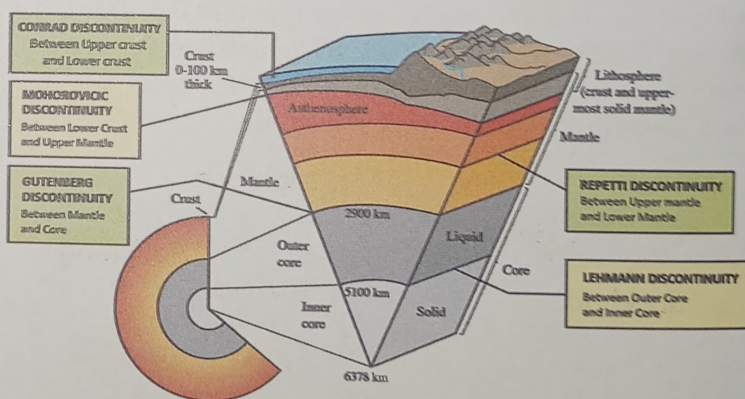
Asthenosphere is not a part of Lithosphere.

Core

- It is made of **Nickel** and **Iron** (NiFe).
- Mean density:** 13 g/cm³.
- Core has two divisions:**
 - Inner core:** It is in **solid form** (1220 km).
 - Outer core:** It is in **liquid form** and shows magnetic properties (2200 km).

By	Crust	Mantle	Core
Volume	1%	84%	15%
Mass	1%	68%	31%

Earth's Discontinuity.

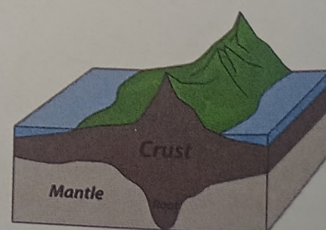


TRICK TO REMEMBER

Mohorovicic → Koi → Conrad
Gutenberg → Red → Repitti
Mujhe → Gulab → Lehmann
Lado

Mountain isostasy

- Mountains rest on the Earth's crust like icebergs floating in water, with deep "roots" extending into the mantle. The higher the mountain, the deeper its root below the Earth's surface to maintain balance.



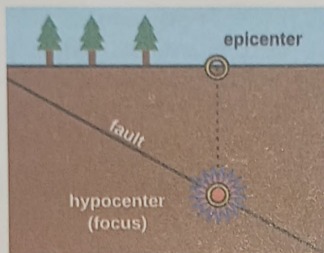
Theory of Plate Tectonics: It is given *Mckenzie* and *Parker* in 1967. This theory suggests that the *Earth's lithosphere* (crust and uppermost mantle) is broken into large, rigid plates that move and interact with each other, driving various geological phenomena such as earthquake.

- Convection current that occurs in the asthenosphere is the force behind the plate movement. The heat from the Earth's interior causes currents of hot rising magma and cooler sinking magma to flow, moving the lithospheric plates along with them.

Earthquake

- An earthquake is an intense shaking of Earth's surface, which is caused by *shifting of Earth's plate*. It is a sudden release of energy along the fault line in the form of waves (also known as *seismic waves*).

- **Epicentre:** It is the point on the Earth's surface directly above the *earthquake's focus* (also known as the *hypocentre*), where the earthquake originates underground.



- **Hypocentre:** It is the point within the Earth where the *earthquake begins*.

- All natural earthquake occur in *lithosphere*.
- Earthquake generate seismic waves, which are vibrations that travel through the earth in the form of *body waves* and *surface waves*.

Body Waves

These waves travel through the Earth's interior with high speed.

They are of two types: *P-waves* and *S-waves*.

<i>P-waves</i>	<i>S-waves</i>
They are also known as primary waves	They are also known as secondary waves
They are the fastest seismic waves (first to record on seismograph) → 7-8 km/sec	They are the slow seismic waves in comparison to P-waves → 4-6 km/sec
They can travel through all medium (solid > liquid > gas)	They can travel through only solids
P-waves are longitudinal compression waves, similar to sound waves	S-waves are transverse waves, that causes the ground to vibrate perpendicular to the direction of the wave
They cause stretching and squeezing because they vibrate parallel to the direction of waves	They create crests and troughs because they vibrate perpendicular to the direction of waves

Note: *Shallow focus, earthquakes have more intensity than the deeper ones.*

Surface Waves

- When body waves come in contact with *surface rocks*, they form surface waves.

- They are more destructive in nature and cause displacement of rocks.

Surface waves are of two types

Love waves

They move the ground side-to-side horizontally.

Rayleigh waves

They produce a complex, rolling motion in the ground (similar to ocean waves).

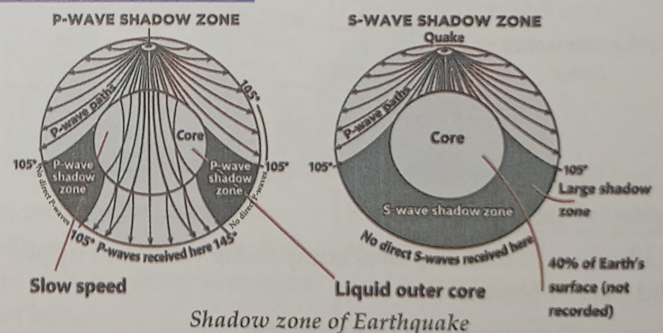
- They are the *last to be recorded* on seismograph.

A seismograph is an instrument that measures and registers seismic waves that travel through the Earth as the outcome of an earthquake.

Scales to Measure Earthquake

<i>Richter Scale</i>	<i>Mercalli Scale</i>
It is an instrument to measure magnitude of the earthquake	It is an instrument to measure the intensity of the earthquake
It measures magnitude from 0 to 10	It measures intensity from 1 to 12
Richter scale relates to the energy released during the earthquake.	Mercalli scale takes into account the visible damage caused by the earthquake.
It is a limitless scale	

Shadow Zone of Waves



- The shadow zone of an earthquake is the area of the Earth, where certain seismic waves are not detected. The shadow zone of *S-waves* (105°-180°) is larger (40% of Earth's surface) than the shadow zone of *P-waves* (105°-145°).

- Animals are sensitive to infrasonic waves, seismic p waves; therefore, they can sense natural calamities like earthquakes, tsunamis beforehand.

Effects of Earthquake

- Tsunami (Japanese term): These are *harbour waves*. It occurs when earthquake occurs at sea floor.

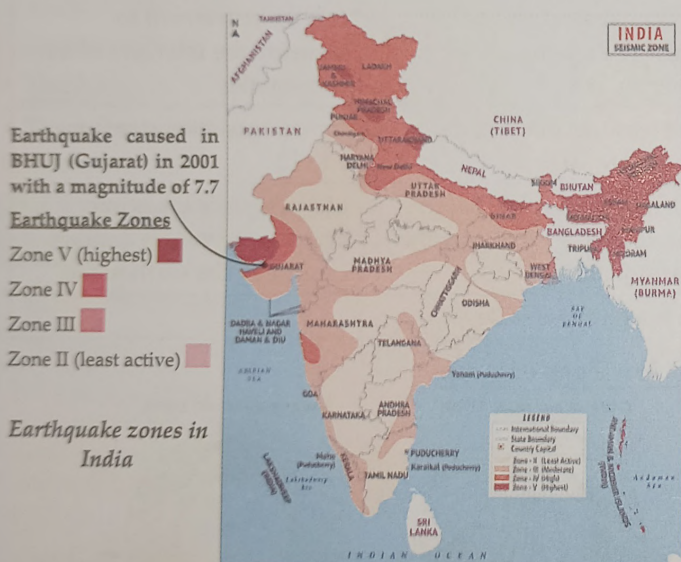
- Soil liquefaction
- Landslide
- Avalanche
- Floods
- Fire

• The Indian Ocean Tsunami of 2004, also known as the "Boxing Day Tsunami", happens to be the deadliest tsunami to date.

• On May 22, 1960, Chile got struck with the strongest earthquake in history (9.5 magnitude), also known as the Great Chilean Earthquake.

Types of Earthquake

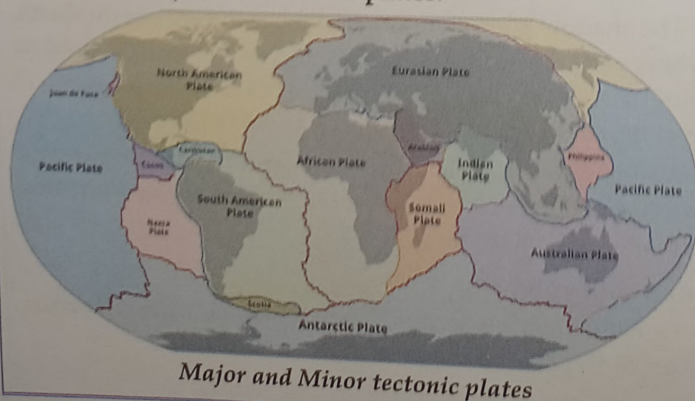
- **Collapse earthquake:** These are small earthquake that occurs when the roof of underground mine or caverns collapse.
- **Explosion earthquake:** An earthquake caused by detonation of a chemical or nuclear device.
- **Tectonic earthquake:** An earthquake caused by movement of earth's tectonic plates.
- **Reservoir induced earthquake:** An occurrence of earthquakes caused by construction of large reservoirs. This is human-induced earthquake. Eg. 1967 Koyna Dam earthquake (Maharashtra).
- **Volcanic earthquake:** An earthquake that occurs as result of volcanic eruption.



Tectonic Plates

- Lithosphere makes plates comprising crust and upper solid part of Mantle.

There are 7 major + few minor plates.



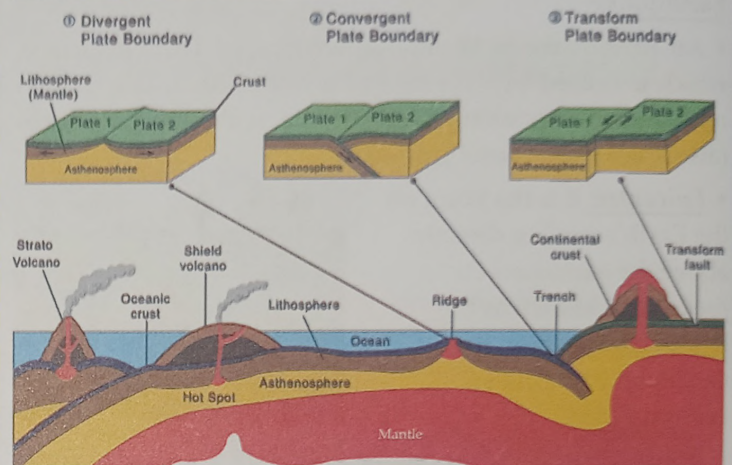
Major Tectonic Plates

- Pacific plate (Largest)
- North American plate
- South American plate
- African plate
- Antarctic plate
- Eurasian plate
- Australian plate

Minor Tectonic Plates

- Cocos plate
- Nazca plate
- Caribbean plate
- Indian plate
- Philippine plate
- Juan de Fuca plate (Smallest)

Three Types of Plate Boundary



Divergent	Convergent	Transform
Occurs when two tectonic plates move away from each other, causes lava to explode, leading to new crust formation.	It happens when plates collide, causing one plate to subduct beneath the other creating a subduction zone, potentially destroying the crust.	It is a plate boundary where plates slide past each other, neither creating nor destroying crust and hence Conservative in nature (San Andreas Fault).

- Divergent plate boundary of two oceanic plates forms mid oceanic ridge, but divergent plate boundary of two continental plates forms Rift valley with block mountains on both sides, like the African Rift Valley.
- At convergent boundaries, dense oceanic crust subducts beneath continental crust, forming trenches and uplifting mountains. Melting of the subducting plate produces magma that rises, creating volcanic chains near the mountains.
- But in case of the Indian plate and Eurasian plate, as both are continental crusts, no major subduction took place, the plates got folded to form the Himalayas (As no subduction occurred here, no volcanoes formed nearby).

Ring of Fire

- It is a result of Convergent Plate Boundary. It is a horse-shoe-shaped region around the Pacific Ocean. It is a high earthquake (90%) and volcano prone region (75%).

FORMATION OF ROCKS

- **Petrology:** The study of rocks is called Petrology.
- Rocks do not have definite composition of mineral constituents. Age of rocks is calculated using Carbon Dating.

Igneous Rocks

- The word "Igneous" derived from the Latin word "Ignis" which means "fire".
- Igneous rocks (also called "Azoic", as no fossils are found in them) are formed through the cooling and solidification of molten rock, known as *magma* or *lava*.

Two Types of Igneous RocksIntrusive Igneous Rocks/
Plutonic Rocks

When the magma cools and solidifies slowly within the Earth's crust, it forms intrusive igneous rocks.

Eg. Granite, Gabbro, etc.

Extrusive Igneous Rocks

When the magma erupts on the Earth's surface as lava, it cools and solidifies quickly. This rapid cooling results in the formation of small crystals or glassy texture.

Eg. Basalt, Pumice (it floats on water), etc.

Plutonic Bodies

Plutonic bodies, also known as plutons, are *large bodies of intrusive igneous rock* that form beneath the Earth's surface.

Types of Plutonic Bodies

Batholiths: A batholith is a large *mass of igneous rock* that forms deep within the Earth's crust and is exposed at the surface through erosion and uplift.

Dykes: Dykes are vertical or *steeply inclined sheets of magma* that cut across the layers of surrounding rock.

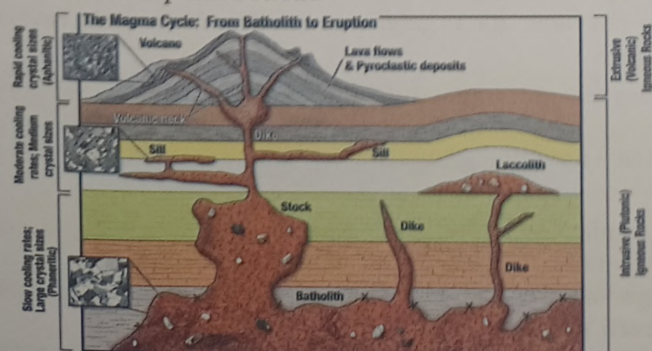
Sills: Sills are horizontal sheets of magma that intrude between layers of pre-existing rock.

Phacolith: A phacolith is a *lens-shaped igneous rock pluton* that forms in folded sedimentary rock found at *Anticlines* and *Synclines*.

Laccoliths: *Dome-shaped intrusions* with a flat base that push overlying layers upward.

It is connected by a *pipe-like structure Conduit*.

Lapoliths: These are *saucer-shaped intrusions* with concave top and bottom.



SEDIMENTARY ROCKS

- Sedimentary rocks are a type of rock that forms from the *accumulation and consolidation of sediments* over time.
- **Lithification** is a process by which loose sediments are transformed into solid sedimentary rock. This occurs through a combination of *compaction and cementation* over long periods of time.
- They exist in layers/strata.

Types of Sedimentary Rocks

1. **Mechanically/Clastically formed:** Sandstone, limestone, conglomerate, shale, loess, etc.
2. **Organically formed:** Chalk, limestone, coal, geyserite, etc.
3. **Chemically formed:** Limestone, halite, chert, etc.

- *Fossils are found in sedimentary rocks.*

METAMORPHIC ROCKS

- Metamorphic rocks are rocks that form when existing rocks (igneous, sedimentary, or even other metamorphic rocks) are subjected to *intense heat, pressure, or volume change*, causing them to undergo physical and chemical changes. This process is also called *Metamorphism*.

- These rocks are formed by recrystallisation and reorganisation of materials within the original rocks.

Two Types of Metamorphic RocksThermal Metamorphism

These are formed due to *sudden temperature change*.

Dynamic Metamorphism

These are formed *without any chemical change*. Generally formed as a result of *high pressure*.

Types of Thermal MetamorphismContact Metamorphism

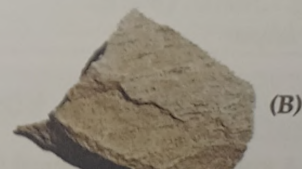
It occurs when rocks are altered by intense heat due to their proximity to magma or lava.

Regional Metamorphism

It occurs when rocks are subjected to *high pressure and temperature*.



(A) Alternate light and dark bands called **Banding**;



(B) Lines formed called **Lineation/foliation**

- *Quartz is not a rock, it is a mineral.*

- *Taj Mahal is made of: Marble (metamorphic rock).*

- *Red Fort is made of: Red sandstone (sedimentary rock).*



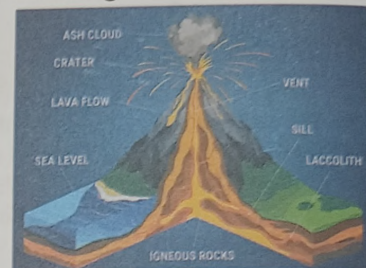
• Granite: Found in Continental crust.
• Basalt: Found in Oceanic crust.



2. **Dormant volcano:** A volcano that has not erupted for a long time but may erupt again in the future.
3. **Extinct volcano:** A volcano that has not erupted for thousands of years and is not expected to erupt again.

Types of Volcano

1. **Cinder cone:** They are **small, steep-sided** volcanoes formed by moderately explosive eruptions and erupts more pyroclastic debris than lava.
2. **Flood Basalt Province:** These volcanoes outpour, highly fluid lava that flows for long distances. Eg. **Deccan Traps** (in West Central region).
3. **Shield:** They are primarily formed by eruption of **basaltic lava** which is **non-viscous fluid**. They become explosive when water enters their vent. Eg. **Kilauea Volcano in Hawaii, USA**.



4. **Composite:** They are generally **cooler and viscous** in nature. They are also known as **Stratovolcano**. Eg. **Mt. Chimborazo and Mt. Fuji**.
5. **Caldera:** It is the **most explosive** type of volcano, it collapses on itself, which means their magma chamber is in close vicinity.
6. **Fissure volcano:** They have an elongated vent or crack formed by divergent plate boundary, through which lava erupts. It doesn't have a hill or mountain-like structure like other volcanoes but shows volcanic activity.
7. **Arc Volcanoes:** Volcanoes formed in the ocean due to convergent plate boundary.

Acidic vs Basic Rocks

Acidic Rocks	Basic Rocks
They are high in silica (mostly feldspar and quartz minerals) upto 80%	They are poor in silica (mostly feldspar and quartz minerals) about 40%
They are light in color	They are dark in color
Due to high silica content, acidic magma cools faster	Due to low silica content, basic magma cools slowly
They are formed from acidic lava (Felsic in nature - high in Silica & Feldspar) which is viscous in nature.	They are formed from basic lava (Mafic in nature - high in Magnesium & Iron) which is fluid in nature.
Eg. Granite, Quartz, Feldspar	Eg. Basalt, Gabbro, Dolerite

Highest volcano: Ojas del Salado (it is located between Chile and Argentina in Andes mountain range).

Lighthouse of Mediterranean: Mount Stromboli

Largest volcano by volume - Mauna Loa, Hawaii.

Volcanic hotspots

- Generally volcanoes are formed between plate boundaries but some volcanoes are formed within the plates and their source of magma is outer core not asthenosphere.

Example: Hawaiian volcanoes like Mauna Loa, Mauna Kia etc are hotspot volcanoes and they're not connected to the Pacific ring of fire.

Geyser

- A geyser is a **hot spring that intermittently erupts steam** and hot water due to geothermal activity beneath the Earth's surface.

- Geysers are commonly found in areas of volcanic activity because the heat from magma or hot rocks below the Earth's surface heats the groundwater, leading to the formation of geysers.

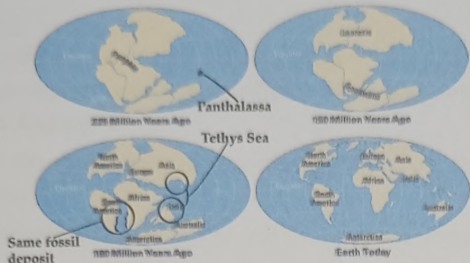
Example: Old faithful geyser in the USA (near this hotspot volcanism can be seen in Yellow Stone National Park), Puga Valley in Ladakh, Manikaran in Himachal Pradesh (at Parvati Valley).

CONTINENTS AND OCEANS

- In 1912, Alfred Wegener gave Continental Drift Theory.
- All of the modern day continents had previously been clumped together in a supercontinent called *Pangaea* and the water body is called *Panthalassa*.

Evidences

- Jig-saw fit.
- Fossil deposits (Paleontology is the study of Fossils).
- Botanical evidences.
- Rocks of same age across the oceans.
- Ghana gold deposit (Placer deposits)- Gold deposits in Ghana were connected to the Brazilian plateau, when two continents laid side by side. As a result, even though the gold deposits are found in Ghana & South African region, the gold veins still remained in South America.
- Gondwana System Sediments: The Gondwana system of sediments (sedimentary rock that is deposited directly by glacial ice, called *Tillite deposits*) from India has counterparts in six other southern landmasses: Africa, Australia, Antarctica, Madagascar, South America, and the Falkland Islands.

Reason for Continental drift

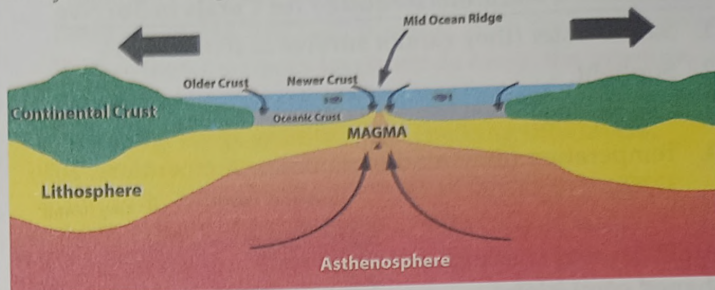
1. Tidal force
 2. Pole fleeing force
- } As suggested by Alfred Wegener
- Actual reason was later given by Arthur Holmes in 1930 (Convection Current Theory) which states that continental drift occurs due to the movement of tectonic plates, which is driven by convection cells in the Earth's mantle.
 - The heat that drives convection cells in the Earth's mantle and plate motion comes mainly from three sources:
 1. Radioactive decay of elements (causing thermal differences).
 2. Residual heat from the Earth's formation about 4.6 billion years ago when gravitational energy was converted into heat.
 3. Latent heat released by the solidification of the Earth's core as it cools down.

Milankovitch Theory

- It describes the collective effects of changes in Earth's movement on its climate over thousands of years.
- In 1596, Abraham Ortelius's work, "Thesaurus Geographicus", first suggested that Earth's continents were not always in their current locations.

SEAFLOOR SPREADING

- Harry H. Hess in 1962, gave Seafloor Spreading Theory.
- Seafloor Spreading is a process by which new oceanic crust is formed at mid-ocean, due to divergent plate boundary. The magma erupts from beneath pushing the older crust away from the ridge in the centre.



- Rocks equidistant from the mid oceanic ridge are of equal age, and their age increases as you move farther away from the ridge.
- Oceanic crust is younger than continental crust.

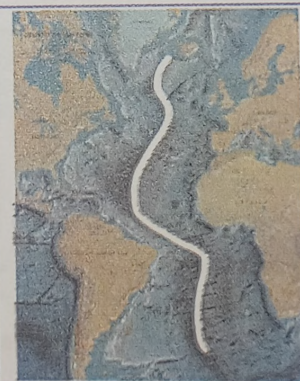
Relief features are of two types:1. Major Relief Features

Four major relief features are there:

Continental Shelf → Continental Slope → Continental Rise
Abyssal plain/Deep sea plains

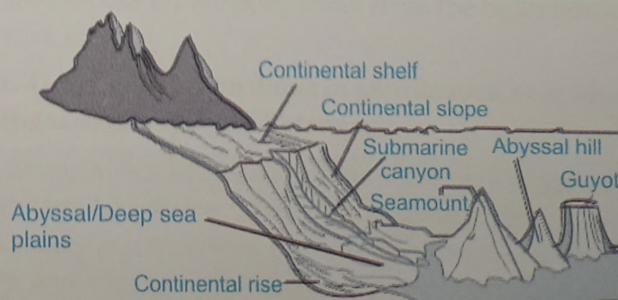
2. Minor Relief Features

Eg. Seamount, Guyot (flat-topped seamount), Mid-Ocean ridge, Trench, Coral, Atoll, etc.



- The Mid-Atlantic Ridge is a mid-ocean ridge located along the floor of the Atlantic Ocean, and part of the longest relief feature in the world.
- It got created due to divergent plate boundary.

- Continental Shelf is the shallowest part of the ocean, having a gradient of 1 degree or less.
- The gradient of the Continental Slope region varies between 2° - 5°.



Relief Features of Ocean

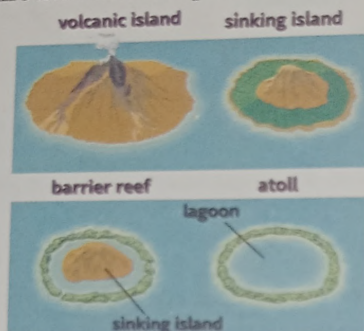
Corals

- Corals are marine organisms known as "*Rainforest of Sea*".
- They exist in symbiotic relationship with *Zooxanthella* algae. The corals secrete CaCO_3 that provides protection to the algae and in return the algae make food for the corals.
- Corals exist in colonies.

Favourable Conditions Required for Corals to Survive

1. Saline water (they cannot survive in fresh water).
2. Sunlight.
3. Clear water.
4. Temperature (they require moderate temperature: $20^\circ\text{--}32^\circ\text{C}$).

Atoll: An atoll is a ring-shaped coral island or reef surrounding a central lagoon, formed from the growth of coral around a sinking volcanic island.



Great Barrier Reef in Australia is the largest barrier reef (colony of corals) in the world.

Coral Bleaching

- When water is *too warm*, corals will expel the algae (*Zooxanthella*) living in their tissues, causing the coral to turn completely white. *Climate change is a major key factor in this change.*

FACTS

- Mariana Trench is the deepest part of the Pacific Ocean (11 km).
- Challenger Deep is the deepest known point in Mariana Trench.
- Atlantic Ocean is the busiest ocean. Sargasso Sea is a unique region in the Atlantic Ocean, which is a borderless sea (brown algae *Sargassum* is seen here).



GEOMORPHOLOGICAL PROCESS

- **Father of geomorphology-** *William Morris Davis*.
- Geomorphology is a scientific discipline, concerned with description and classification of earth's topographic features. All geomorphologic process occur due to two types of forces: **Endogenic Forces (internal)** and **Exogenic Forces (external)**.

Endogenic Processes

- These are geological processes that occur due to endogenic forces. All relief features on earth are subject to endogenic and exogenic forces. **Example-**
 - **Himalayas:** They are continuously **increasing (endogenic forces > exogenic forces)**.
 - **Aravali:** They are continuously **decreasing (exogenic forces are > endogenic forces)**.
- • Endogenic forces get their energy from within the earth (recall Convection Current Theory).

Effects of Endogenic Forces

1. **Diastrophism:** It is a process of **deformation and movement** of Earth's crust. These movements are responsible for shaping large-scale landforms over geological time periods.

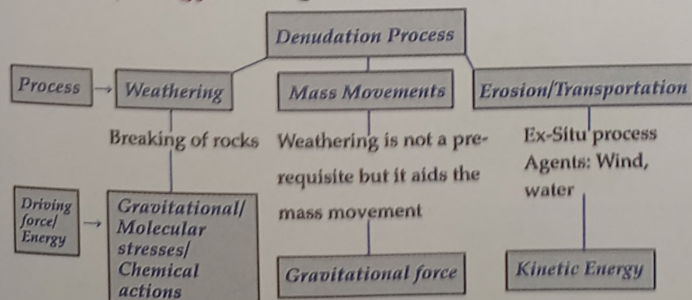
Types of Diastrophism

- a) **Orogenic Process:** It is a process through which mountains are built.
- b) **Epeirogenic Process:** It involves upliftment or warping of large parts of the Earth's crust.
- c) **Earthquake:** It involves shaking of Earth.
- d) **Plate Tectonics:** It explains how major landforms are created as a result of Earth's subterranean movements.

2. **Volcanoes:** They are commonly described as openings/vents where lava or magma erupts.

Exogenic Processes

- These are geological processes that occur on the Earth's surface due to external forces (exogenic forces).
- **Denudation** refers to long-term processes that cause the wearing away of earth's surface, leading to a reduction in elevation and relief of landforms and landscapes.
- **Exogenic agents are:** Wind, glaciers, water.
- **Source of energy** for exogenic forces is the Sun.



Weathering

- Weathering is the process of rocks and minerals breaking down or dissolving on the Earth's surface.

Types of Weathering

1. **Physical Weathering:** This involves the breakdown of rocks and soils **through direct contact** with atmospheric conditions, such as **heat, water, ice, and pressure**.

Examples-

Unloading and Expansion- When rocks deep inside the Earth are under very high pressure from the layers above them. When the upper layers erode away, the pressure decreases and the rock expands and cracks, forming sheets. Eg. **Large granite rocks cracking into layers**.

Frost Wedging- Water enters small cracks in rocks. When temperature drops, the water freezes and expands. This expansion pushes the rock apart, and repeated freezing and thawing eventually breaks the rock. **Common in: Cold mountainous areas**.

Salt Crystallisation- Salt water enters tiny cracks in rocks. When the water evaporates, salt crystals remain and grow inside the cracks. The growing crystals exert pressure and break the rock apart. **Common in: Coastal areas and deserts**.

2. **Chemical Weathering:** This involves the decomposition of rocks and soils **through chemical reactions**, such as **oxidation and reduction, carbonation, solution, and hydration**.

Solution: Occurs mainly in soft rocks which are made up of soluble minerals Eg. Nitrates, Sulfates, Potassium, and these get dissolved in water or acid or leaching takes place.

Eg. solution process can be seen in limestone where the CaCO_3 gets dissolved.

Hydration: Occurs in rocks made up of minerals that absorb water and expand, causing the rock to weaken or change into a new mineral.

3. **Biological Weathering:** It is also known as **organic weathering**, is the breakdown of rocks and soils through the activities of living organisms such as earthworms, rodents, etc.

Weathering is an in-situ process.

Effects of Weathering

1. **Exfoliation:** It is a process when large, curved plates, or slabs of rocks are stripped away from the outer surface of a rock mass.

Eg. a large exfoliation dome in granite rock near Bhongir (Bhuvanagiri) town in Telangana.

Tor: Rounded hill like structure formed due to physical weathering (exfoliation), generally in granite rock.

[The exfoliation first turns a giant rock to an exfoliation dome, and after sometimes the rate of exfoliation declines, giving it a stable hill like structure that's Tor]